How to Set Up Hadoop Cluster with HDFS High Availability With Hadoop And Zookeeper With Hbase

Hadoop
Hadoop is a framework that allows you to store and process large amounts of data across clusters of computers using simple programming models. It is designed to scale from a single server to thousands of machines.

Key Components:
HDFS (Hadoop Distributed File System): Stores data across multiple machines.
MapReduce: Processes data in parallel across the cluster.
YARN (Yet Another Resource Negotiator): Manages resources and schedules tasks.

Why use it?
It's great for big data because it's scalable, fault-tolerant, and cost-effective.

Zookeeper
Zookeeper is a centralized service for managing distributed systems. It helps in maintaining configuration information, naming, synchronization, and group services.

Why use it?
It ensures that all the nodes in a distributed system work together smoothly and consistently. It's like a traffic controller for distributed systems.

Key Features:
Reliability: Keeps the system running even if some nodes fail.
Consistency: Ensures all nodes have the same view of the system.
Coordination: Helps in leader election, locking, and synchronization.

ZKFC (ZooKeeper Failover Controller)
ZKFC is a component used in Hadoop's High Availability (HA) setup for NameNode. It ensures that if the active NameNode fails, a standby NameNode takes over quickly.
How it works:
ZKFC monitors the health of the NameNode.
If the active NameNode fails, ZKFC uses Zookeeper to elect a new active NameNode.
This ensures no single point of failure in the Hadoop cluster.

Why use it?
It provides automatic failover for the NameNode, ensuring the Hadoop cluster remains available even during failures.

HBASE
HBase is an open-source, distributed, and scalable NoSQL database that is designed to store and manage large amounts of data across many machines. It is modeled after Google's Bigtable and is built on top of Hadoop, which allows it to handle big data efficiently.

Here's a simple breakdown:

1. **Data Storage**: HBase stores data in tables, similar to a relational database, but it doesn't use SQL. Each table consists of rows and columns.
2. **Scalability**: It can scale to handle petabytes of data across many servers.
3. **Real-time Access**: HBase allows quick, real-time access to data, making it suitable for applications that need fast read/write operations.
4. **Distributed**: The data is distributed across multiple machines, which allows it to handle large datasets.
5. **Used for Big Data**: It is commonly used in big data applications like web indexing, data warehousing, or storing large logs.

In short, HBase is great for storing huge amounts of structured data that need to be accessed quickly and reliably.

10.X.X.1 → NameNode (Active namenode)
10.X.X.2 → NameNode (Stnadby namenode)
10.X.X.3 → DataNode

//Prerequisites (ALL nodes)
Access to a terminal window/command line.
Sudo or root privileges on local/remote machines.
Ubuntu version 20.04, 22.04, or 24.04 installed.

```
//Install JDK on Ubuntu (ALL nodes)
sudo apt update
sudo apt install openjdk-8-jdk -y
java -version; javac -version
sudo apt install openssh-server openssh-client -y


//Create Hadoop User (ALL nodes)
sudo adduser sysadmin
su - sysadmin
ssh-keygen -t rsa -P '' -f ~/.ssh/id_rsa
cat ~/.ssh/id_rsa.pub >> ~/.ssh/authorized_keys
chmod 0600 ~/.ssh/authorized_keys
ssh localhost



cat ~/.ssh/id_rsa.pub (for all 3 node)

ssh-rsa
AAAAB3NzaC1yc2EAAAADAQABAAABgQCQ1+XWiyQwj+DWkK2//74x8sAxoK757W7TgG
0xFb6epY8Gj82Y3HHgt9Q3cuIo4lEdowSUYhp1ZVTwDe0KaqBGQK56OCtoJAtjNNz28SnP287
+GRUWFh6maTov1kgRi1zIM6A4ZonEP20cGaMPrBKO2vFLq2mT2q4YdLkJpQJ8PxAyLncytn0
sysadmin@worker-1
```

ssh-rsa
AAAAB3NzaC1yc2EAAAADAQABAAABgQCoF6MCdo19qNDUv9oUmoHBHTXvYMVxAcG
z8QKbOe90/3CJjN5Bbev8rrbw34g84tQV+ewqYsA9wAsUHK8XaeQtLvuU+htq36S5Zhs9GlDwc
kpFRrz7KnfEIcm2k3vham4u4XiVA9kiKZGCEkaR4q0K9qvcR62M= sysadmin@master-2

ssh-rsa
AAAAB3NzaC1yc2EAAAADAQABAAABgQCQ+d0i2Y7X5YuXuJZC+jq8IWe0+VLfeSg7aHG
KfqqehQnmcLhs/c7IcQyc2l362b9FdfBjE/gMVW+lTG8zQl3AfVJfz5SR/
mlLxOwTd+cN6mjdwhmkHmR42TABwffWvrIBcm0xk9047He3bzbogBhtpxxC10I+VDBiYUHXj
JlIIm1OvYN0hevba4Ov46sl9DHCoAic5Akj3iUJFQLp7gzDd9lgBibRb4bx8Hfq0unPWJsAubwPlT
izPMYm6R/jrpIfIU= sysadmin@master

// Copy above .pub key this in all nodes in
vim .ssh/authorized_keys

check passwdless ssh is happen or not
ssh sysadmin@10.X.X.2
ssh sysadmin@10.X.X.1
ssh sysadmin@10.X.X.3

Add Hostname in All nodes

vim /etc/hosts

10.X.X.1 master-1
10.X.X.2 master-2
10.X.X.3 worker-1

Note: Do it below steps on Datanode then You can Scp all files and folders to other 2 nodes

1. Download and Extract Hadoop and Zookeeper Binaries
Hadoop:
cd *home*/sysadmin
sudo wget https://archive.apache.org/dist/hadoop/common/hadoop-3.1.1/hadoop-3.1.1.tar.gz
sudo tar -xvzf hadoop-3.1.1.tar.gz
sudo chown -R sysadmin:sysadmin hadoop-3.1.1
sudo chmod -R 777 hadoop-3.1.1

Make File System Setup (On Each Node)

sudo mkdir -p /home/sysadmin/dfs/name
sudo mkdir -p /home/sysadmin/dfs/data
sudo mkdir -p /home/sysadmin/dfs/journal
sudo chown -R sysadmin:sysadmin /home/sysadmin/dfs
sudo chmod -R 777 /home/sysadmin/dfs

2. Configure Hadoop and Zookeeper

Hadoop Configuration Files (All nodes)

 # vim /home/sysadmin/hadoop-3.1.1/etc/hadoop/core-site.xml

```
<?xml version="1.0" encoding="UTF-8"?>
<configuration>
  <property>
    <name>fs.defaultFS</name>
    <value>hdfs://mycluster</value>
    <final>true</final>
  </property>

  <property>
    <name>io.file.buffer.size</name>
    <value>131072</value>
    <description>Buffer size</description>
  </property>

  <property>
    <name>dfs.journalnode.edits.dir</name>
    <value>/home/sysadmin/dfs/journal</value>
    <description>journal data dir</description>
  </property>

  <property>
    <name>hadoop.security.authentication</name>
    <value>simple</value>
   </property>

   <property>
    <name>hadoop.security.authorization</name>
    <value>false</value>
   </property>

   <property>
    <name>fs.trash.interval</name>
    <value>360</value>
   </property>

   <property>
    <name>ha.failover-controller.active-standby-elector.zk.op.retries</name>
    <value>120</value>
   </property>

   <property>
    <name>dfs.ha.nn.not-become-active-in-safemode</name>
    <value>true</value>
   </property>
```

```
    <property>
      <name>hadoop.http.cross-origin.allowed-headers</name>
      <value>X-Requested-With,Content-Type,Accept,Origin,WWW-Authenticate,Accept-
Encoding,Transfer-Encoding</value>
    </property>

    <property>
      <name>hadoop.http.cross-origin.allowed-methods</name>
      <value>GET,PUT,POST,OPTIONS,HEAD,DELETE</value>
    </property>

    <property>
      <name>hadoop.http.cross-origin.allowed-origins</name>
      <value>*</value>
    </property>

    <property>
      <name>hadoop.http.cross-origin.max-age</name>
      <value>1800</value>
    </property>

   <property>
      <name>hadoop.proxyuser.sysadmin.groups</name>
      <value>*</value>
   </property>

   <property>
      <name>hadoop.proxyuser.sysadmin.hosts</name>
      <value>*</value>
   </property>

   <property>
      <name>dfs.namenode.accesstime.precision</name>
      <value>3600000</value>
   </property>

   <property>
      <name>nfs.superuser</name>
      <value>sysadmin</value>
   </property>

   <property>
      <name>nfs.export.point</name>
      <value>/</value>
   </property>
<property>
   <name>hs.zookeeper.quorum</name>
   <value>10.X.X.1:2181,10.X.X.2:2181,10.X.X.3:2181</value>
</property>
   <property>
      <name>hadoop.tmp.dir</name>
      <value>/home/sysadmin/hadoop/tmp</value>
```

```
  </property>

</configuration>
```

 # vim /usr/local/hadoop/etc/hadoop/hdfs-site.xml

```
<?xml version="1.0" encoding="UTF-8"?>
<configuration>

  <!-- Enable Block Access Token for Security -->
  <property>
    <name>dfs.block.access.token.enable</name>
    <value>true</value>
  </property>

  <!-- Initial Delay for Block Report -->
  <property>
    <name>dfs.blockreport.initialDelay</name>
    <value>120</value>
  </property>

  <!-- Default Block Size -->
  <property>
    <name>dfs.blocksize</name>
    <value>134217728</value>
  </property>

  <!-- Enable Short-Circuit Local Reads -->

  <!-- Disable Client Retry Policy -->
  <property>
    <name>dfs.client.retry.policy.enabled</name>
    <value>false</value>
  </property>

  <!-- Content Summary Limit -->
  <property>
    <name>dfs.content-summary.limit</name>
    <value>5000</value>
  </property>

  <!-- DataNode Balance Bandwidth -->
  <property>
    <name>dfs.datanode.balance.bandwidthPerSec</name>
    <value>6250000</value>
  </property>

  <!-- DataNode Reserved Space -->
  <property>
    <name>dfs.datanode.du.reserved</name>
    <value>61498813440</value>
  </property>
```

```
<!-- Tolerate No Failed Volumes -->
<property>
  <name>dfs.datanode.failed.volumes.tolerated</name>
  <value>0</value>
  <final>true</final>
</property>

<!-- Heartbeat Interval -->
<property>
  <name>dfs.heartbeat.interval</name>
  <value>3</value>
</property>

<!-- HTTP Policy -->
<property>
  <name>dfs.http.policy</name>
  <value>HTTP_ONLY</value>
</property>

<!-- Maximum Replication -->
<property>
  <name>dfs.replication.max</name>
  <value>50</value>
</property>

<!-- Enable WebHDFS -->
<property>
  <name>dfs.webhdfs.enabled</name>
  <value>true</value>
  <final>true</final>
</property>

<!-- Max Transfer Threads -->
<property>
  <name>dfs.datanode.max.transfer.threads</name>
  <value>4096</value>
</property>

<!-- NameNode and DataNode Directories -->
<property>
  <name>dfs.namenode.name.dir</name>
  <value>/home/sysadmin/dfs/name</value>
</property>

<property>
  <name>dfs.datanode.data.dir</name>
  <value>/home/sysadmin/dfs/data</value>
</property>

<!-- Default Replication Factor -->
<property>
```

```
  <name>dfs.replication</name>
  <value>2</value>
</property>

<!-- Disable Permissions -->
<property>
  <name>dfs.permissions</name>
  <value>false</value>
</property>

<!-- High Availability Configuration -->
<property>
  <name>dfs.nameservices</name>
  <value>mycluster</value>
</property>

<property>
  <name>dfs.ha.automatic-failover.enabled</name>
  <value>true</value>
</property>

<property>
  <name>dfs.ha.namenodes.mycluster</name>
  <value>nn1,nn2</value>
</property>

<!-- NameNode RPC and HTTP Addresses -->
<property>
  <name>dfs.namenode.rpc-address.mycluster.nn1</name>
  <value>10.X.X.1:8020</value>
</property>

<property>
  <name>dfs.namenode.rpc-address.mycluster.nn2</name>
  <value>10.X.X.2:8020</value>
</property>

<property>
  <name>dfs.namenode.http-address.mycluster.nn1</name>
  <value>10.X.X.1:9870</value>
</property>

<property>
  <name>dfs.namenode.http-address.mycluster.nn2</name>
  <value>10.X.X.2:9870</value>
</property>

<!-- JournalNode Shared Edits Directory -->
<property>
  <name>dfs.namenode.shared.edits.dir</name>
  <value>qjournal://10.X.X.2:8485; 10.X.X.1:8485; 10.X.X.3:8485/mycluster</value>
</property>
```

```
<property>
  <name>dfs.journalnode.edits.dir</name>
  <value>/home/sysadmin/dfs/journal</value>
</property>

<!-- Failover Proxy Configuration -->
<property>
  <name>dfs.client.failover.proxy.provider.mycluster</name>
  <value>org.apache.hadoop.hdfs.server.namenode.ha.ConfiguredFailoverProxyProvider</value>
</property>

<!-- ZooKeeper Configuration -->
<property>
  <name>ha.zookeeper.quorum</name>
  <value>10.X.X.2:2181, 10.X.X.1:2181, 10.X.X.3:2181</value>
</property>

<!-- Fencing Configuration -->
<property>
  <name>dfs.ha.fencing.methods</name>
  <value>shell(/bin/true)</value>
</property>

<property>
  <name>dfs.ha.fencing.ssh.private-key-files</name>
  <value>/home/sysadmin/.ssh/id_rsa</value>
</property>

<!-- Hadoop Proxy Users -->
<property>
  <name>hadoop.proxyuser.sysadmin.groups</name>
  <value>*</value>
</property>

<property>
  <name>hadoop.proxyuser.sysadmin.hosts</name>
  <value>*</value>
</property>

<!-- Other HDFS Configuration -->
<property>
  <name>dfs.namenode.accesstime.precision</name>
  <value>3600000</value>
</property>

<property>
  <name>dfs.namenode.audit.log.async</name>
  <value>true</value>
</property>

<property>
```

```
    <name>dfs.namenode.audit.log.token.tracking.id</name>
    <value>true</value>
  </property>

  <property>
    <name>dfs.namenode.audit.loggers</name>
    <value>default</value>
  </property>

</configuration>
```

 # vim /usr/local/hadoop/etc/hadoop/yarn-site.xml (optional)

```
<configuration>
   <property>
      <name>yarn.resourcemanager.ha.enabled</name>
      <value>true</value>
   </property>
   <property>
      <name>yarn.resourcemanager.cluster-id</name>
      <value>yarn-cluster</value>
   </property>
   <property>
      <name>yarn.resourcemanager.ha.rm-ids</name>
      <value>rm1,rm2</value>
   </property>
   <property>
      <name>yarn.resourcemanager.hostname.rm1</name>
      <value>10.X.X.1</value>
   </property>
   <property>
      <name>yarn.resourcemanager.hostname.rm2</name>
      <value>10.X.X.2</value>
   </property>
</configuration>
```

 # vim /usr/local/hadoop/etc/hadoop/mapred-site.xml (optional)

```
<configuration>
   <property>
      <name>mapreduce.framework.name</name>
      <value>yarn</value>
   </property>
</configuration>
```

scp *-r hadoop-3.1.1* sysadmin@10.X.X.1:/home/sysadmin/
scp *-r hadoop-3.1.1* sysadmin@10.X.X.2:/home/sysadmin/

3. Zookeeper Configuration (All nodes)

cd *home*/sysadmin
sudo wget https://archive.apache.org/dist/zookeeper/zookeeper-3.4.6/zookeeper-3.4.6.tar.gz
sudo tar -xvzf zookeeper-3.4.6.tar.gz
sudo chown -R sysadmin:sysadmin zookeeper-3.4.6
sudo chmod -R 777 zookeeper-3.4.6

1. vim /usr/local/zookeeper/conf/zoo.cfg

tickTime=2000
initLimit=10
syncLimit=5
dataDir=/home/sysadmin/zookeeper-3.4.6/data
clientPort=2181
clientPortAddress=10.X.X.1 #10.X.X.2 #10.X.X.3
server.1=10.X.X.1:2888:3888
server.2=10.X.X.2:2888:3888
server.3=10.X.X.3:2888:3888
4lw.commands.whitelist=stat, ruok, conf, isro, srvr

echo "3" > /home/sysadmin/zookeeper-3.4.6/data/myid (10.X.X.3)
echo "2" > /home/sysadmin/zookeeper-3.4.6/data/myid (10.X.X.2)
echo "1" > /home/sysadmin/zookeeper-3.4.6/data/myid (10.X.X.1)

scp home/sysadmin/zookeeper-3.4.6 sysadmin@10.X.X.1:/home/sysadmin/
scp home/sysadmin/zookeeper-3.4.6 sysadmin@10.X.X.2:/home/sysadmin/

Do the chnages in zoo.cfg & myid with respest IP

ZooKeeper Test (on any machine):

echo "ruok" | nc 10.X.X.2 2181

4. Entry in .bashrc For All Nodes

vim ~/.bashrc

export HADOOP_HOME=/home/sysadmin/hadoop-3.1.1
export PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:$HADOOP_HOME/bin:
$HADOOP_HOME/sbin
export HADOOP_CONF_DIR=/home/sysadmin/hadoop-3.1.1/etc/hadoop
export HDFS_NAMENODE_USER=sysadmin
export HDFS_DATANODE_USER=sysadmin
export HDFS_SECONDARYNAMENODE_USER=sysadmin
export JAVA_HOME=/usr/lib/jvm/java-8-openjdk-amd64

export PATH=$JAVA_HOME/bin:$PATH
export HADOOP_MAPRED_HOME=/home/sysadmin/hadoop-3.1.1
export HADOOP_COMMON_HOME=/home/sysadmin/hadoop-3.1.1
export HADOOP_HDFS_HOME=/home/sysadmin/hadoop-3.1.1
export ZOOKEEPER_HOME=/home/sysadmin/zookeeper-3.4.6
export ZOOKEEPER_LOG_DIR=$ZOOKEEPER_HOME/logs
export PATH=$PATH:$JAVA_HOME/bin:$HADOOP_HOME/bin:$HADOOP_HOME/sbin:$ZOOKEEPER_HOME/bin
export PATH=$PATH:$JAVA_HOME/bin:$HADOOP_HOME/bin:$HADOOP_HOME/sbin:$HBASE_HOME/bin
export HADOOP_CONF_DIR=/home/sysadmin/hadoop-3.1.1/etc/hadoop/
export HADOOP_LOG_DIR=$HADOOP_HOME/logs

source ~/.bashrc

scp ~/.bashrc sysadmin@10.X.X.1:/home/sysadmin/
scp ~/.bashrc sysadmin@10.X.X.2:/home/sysadmin/

5. Start the cluster

// start journalnode
 on all nodes (from namenode standby datanode)
hadoop-daemon.sh start journalnode

// On active namenode
hdfs namenode -format

```
2025-03-19 12:09:12,004 INFO util.GSet: Computing capacity for map INodeMap
2025-03-19 12:09:12,004 INFO util.GSet: VM type       = 64-bit
2025-03-19 12:09:12,006 INFO util.GSet: 1.0% max memory 2.6 GB = 26.4 MB
2025-03-19 12:09:12,007 INFO util.GSet: capacity      = 2^22 = 4194304 entries
2025-03-19 12:09:12,011 INFO namenode.FSDirectory: ACLs enabled? false
2025-03-19 12:09:12,011 INFO namenode.FSDirectory: POSIX ACL inheritance enabled? true
2025-03-19 12:09:12,012 INFO namenode.FSDirectory: XAttrs enabled? true
2025-03-19 12:09:12,012 INFO namenode.NameNode: Caching file names occurring more than 10 times
2025-03-19 12:09:12,050 INFO snapshot.SnapshotManager: Loaded config captureOpenFiles: false, skipCaptureAccessTimeOnlyChange: false, snapshot
DiffAllowSnapRootDescendant: true, maxSnapshotLimit: 65536
2025-03-19 12:09:12,063 INFO snapshot.SnapshotManager: SkipList is disabled
2025-03-19 12:09:12,070 INFO util.GSet: Computing capacity for map cachedBlocks
2025-03-19 12:09:12,070 INFO util.GSet: VM type       = 64-bit
2025-03-19 12:09:12,070 INFO util.GSet: 0.25% max memory 2.6 GB = 6.6 MB
2025-03-19 12:09:12,070 INFO util.GSet: capacity      = 2^20 = 1048576 entries
2025-03-19 12:09:12,134 INFO metrics.TopMetrics: NNTop conf: dfs.namenode.top.window.num.buckets = 10
2025-03-19 12:09:12,135 INFO metrics.TopMetrics: NNTop conf: dfs.namenode.top.num.users = 10
2025-03-19 12:09:12,135 INFO metrics.TopMetrics: NNTop conf: dfs.namenode.top.windows.minutes = 1,5,25
2025-03-19 12:09:12,152 INFO namenode.FSNamesystem: Retry cache on namenode is enabled
2025-03-19 12:09:12,153 INFO namenode.FSNamesystem: Retry cache will use 0.03 of total heap and retry cache entry expiry time is 600000 millis
2025-03-19 12:09:12,163 INFO util.GSet: Computing capacity for map NameNodeRetryCache
2025-03-19 12:09:12,163 INFO util.GSet: VM type       = 64-bit
2025-03-19 12:09:12,165 INFO util.GSet: 0.029999999329447746% max memory 2.6 GB = 809.9 KB
2025-03-19 12:09:12,165 INFO util.GSet: capacity      = 2^17 = 131072 entries
2025-03-19 12:09:16,047 INFO namenode.FSImage: Allocated new BlockPoolId: BP-1012140633-10.13.10.217-1742366356046
2025-03-19 12:09:16,208 INFO common.Storage: Storage directory /home/sysadmin/dfs/name has been successfully formatted.
2025-03-19 12:09:16,814 INFO namenode.FSImageFormatProtobuf: Saving image file /home/sysadmin/dfs/name/current/fsimage.ckpt_000000000000000000
0 using no compression
2025-03-19 12:09:17,095 INFO namenode.FSImageFormatProtobuf: Image file /home/sysadmin/dfs/name/current/fsimage.ckpt_0000000000000000000 of si
ze 393 bytes saved in 0 seconds .
2025-03-19 12:09:17,146 INFO namenode.NNStorageRetentionManager: Going to retain 1 images with txid >= 0
2025-03-19 12:09:17,242 INFO namenode.NameNode: SHUTDOWN_MSG:
/************************************************************
SHUTDOWN_MSG: Shutting down NameNode at master-2/10.13.10.217
************************************************************/
sysadmin@master-2:~/hadoop-3.1.1$
```

hdfs --daemon start namenode

// On standby
hdfs namenode -bootstrapStandby

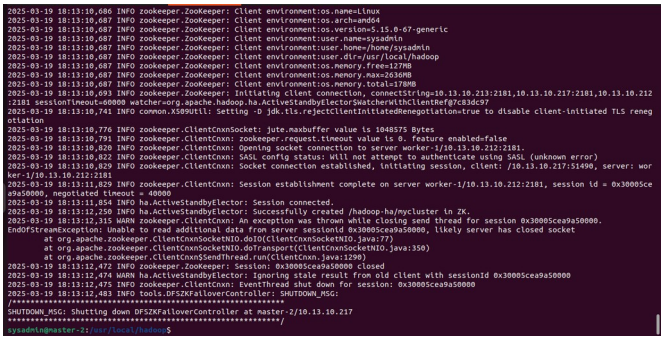
```
2025-03-19 12:10:14,318 INFO namenode.NameNode: createNameNode [-bootstrapStandby]
2025-03-19 12:10:14,698 INFO ha.BootstrapStandby: Found nn: nn1, ipc: master-2/10.13.10.217:8020
2025-03-19 12:10:15,223 WARN common.Util: Path /home/sysadmin/dfs/name should be specified as a URI in configuration files. Please update hdfs
 configuration.
2025-03-19 12:10:15,241 WARN common.Util: Path /home/sysadmin/dfs/name should be specified as a URI in configuration files. Please update hdfs
 configuration.
2025-03-19 12:10:16,997 INFO ipc.Client: Retrying connect to server: master-2/10.13.10.217:8020. Already tried 0 time(s); retry policy is Retr
yUpToMaximumCountWithFixedSleep(maxRetries=10, sleepTime=1000 MILLISECONDS)
=====================================================
About to bootstrap Standby ID nn2 from:
           Nameservice ID: mycluster
        Other Namenode ID: nn1
  Other NN's HTTP address: http://10.13.10.217:9870
  Other NN's IPC  address: master-2/10.13.10.217:8020
             Namespace ID: 1403625508
            Block pool ID: BP-1012140633-10.13.10.217-1742366356046
               Cluster ID: CID-200f4de7-e41a-4239-abb2-ae605fa2e405
           Layout version: -64
       isUpgradeFinalized: true
=====================================================
2025-03-19 12:10:17,475 INFO common.Storage: Storage directory /home/sysadmin/dfs/name has been successfully formatted.
2025-03-19 12:10:17,545 WARN common.Util: Path /home/sysadmin/dfs/name should be specified as a URI in configuration files. Please update hdfs
 configuration.
2025-03-19 12:10:17,546 WARN common.Util: Path /home/sysadmin/dfs/name should be specified as a URI in configuration files. Please update hdfs
 configuration.
2025-03-19 12:10:17,651 INFO namenode.FSEditLog: Edit logging is async:true
2025-03-19 12:10:17,746 INFO namenode.TransferFsImage: Opening connection to http://10.13.10.217:9870/imagetransfer?getimage=1&txid=0&storageI
nfo=-64:1403625508:1742366356046:CID-200f4de7-e41a-4239-abb2-ae605fa2e405&bootstrapstandby=true
2025-03-19 12:10:17,939 INFO common.Util: Combined time for file download and fsync to all disks took 0.01s. The file download took 0.01s at 0
.00 KB/s. Synchronous (fsync) write to disk of /home/sysadmin/dfs/name/current/fsimage.ckpt_0000000000000000000 took 0.00s.
2025-03-19 12:10:17,940 INFO namenode.TransferFsImage: Downloaded file fsimage.ckpt_0000000000000000000 size 393 bytes.
2025-03-19 12:10:18,096 INFO namenode.NameNode: SHUTDOWN_MSG:
/************************************************************
SHUTDOWN_MSG: Shutting down NameNode at master-1/10.13.10.213
************************************************************/
sysadmin@master-1:~/hadoop-3.1.1$
```

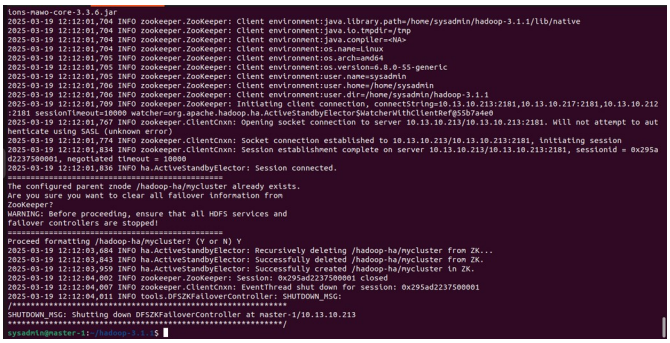
hdfs --daemon start namenode

// start Zookeeper on all nodes (from namenode standby datanode)
zkServer.sh start
zkServer.sh status

// Start Datanode
hadoop-daemon.sh start datanode

// On Active & then Standby
hdfs zkfc -formatZK
hdfs --daemon start zkfc



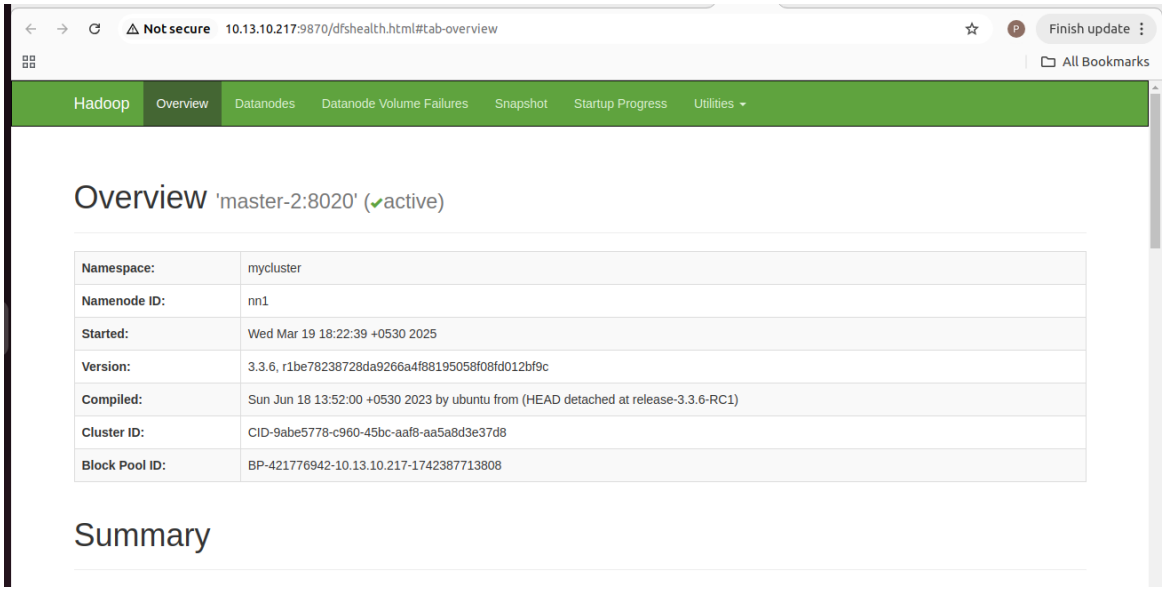


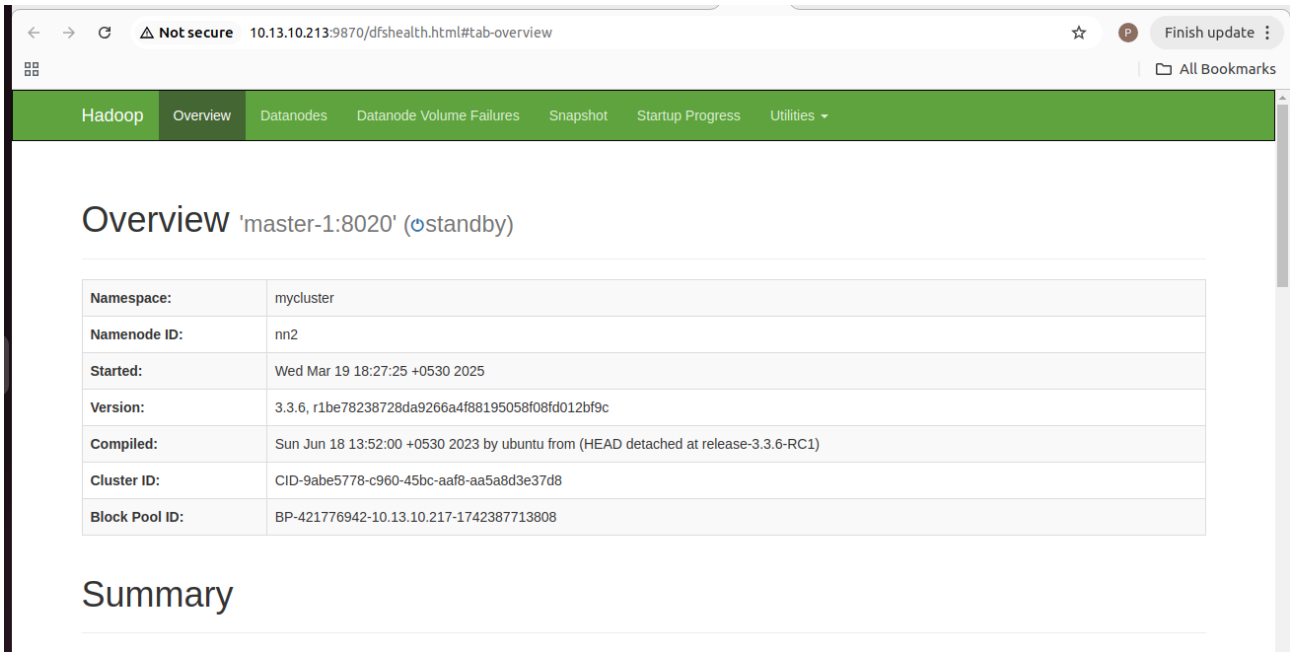
// Start ResourceManager (Optional)
/usr/local/hadoop/bin/yarn --daemon start resourcemanager

// Start NodeManager (All Nodes) (optional)
/usr/local/hadoop/bin/yarn --daemon start nodemanager

6. Check On terminal in All nodes using below command.
jps

7. Verify the cluster On UI
http://10.X.X.1:9870/ ---> Active Namenode
http://10.X.X.2:9870/ ---> Standby Namenode





8. Check Automatic Failover

With the above setup, if the Active NameNode (10.X.X.1) fails, the Standby NameNode (10.X.X.2) will automatically take over as the Active NameNode.

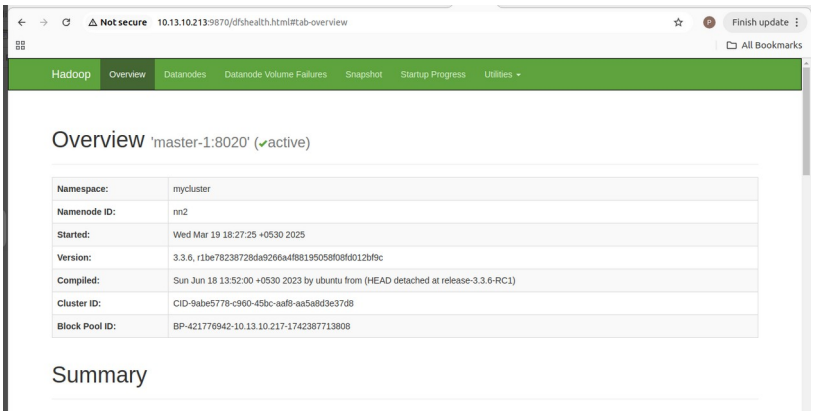
In the Active namenode, kill the namenode daemon to change the Standby name node to active namenode.

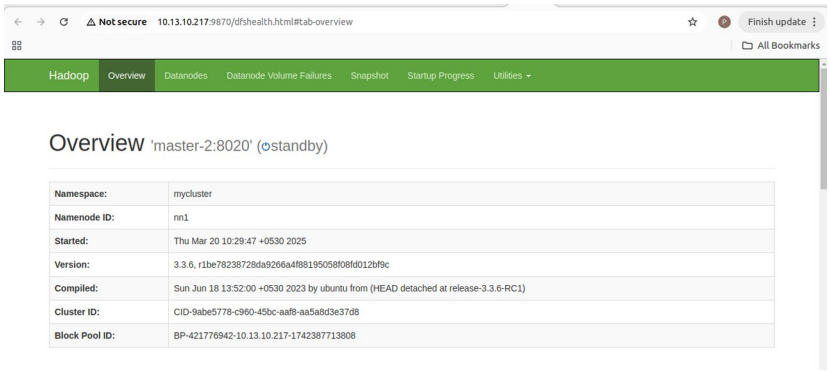
Jps

sudo kill -9 <namenode process ID> Or stop Namenode

```
sysadmin@master-2:~$ jps
214787 JournalNode
215589 NameNode
225851 Jps
215385 DFSZKFailoverController
215230 QuorumPeerMain
sysadmin@master-2:~$ hadoop-daemon.sh stop namenode
WARNING: Use of this script to stop HDFS daemons is deprecated.
WARNING: Attempting to execute replacement "hdfs --daemon stop" instead.
sysadmin@master-2:~$ jps
225907 Jps
214787 JournalNode
215385 DFSZKFailoverController
215230 QuorumPeerMain
sysadmin@master-2:~$
```

Open the two nodes through web browser and check the status.





Install and Configure HBase

Run the following on all machines:

```
sudo wget https://archive.apache.org/dist/hbase/2.3.5/hbase-2.3.5-bin.tar.gz
sudo tar -xzf hbase-2.3.5-bin.tar.gz
sudo chown -R sysadmin:sysadmin hbase-2.3.5
sudo chmod -R 777 hbase-2.3.5
```

1. sudo vim/home/sysadmin/hbase-2.3.5/conf/hbase-site.xml

```
<?xml version="1.0"?>
<configuration>
  <!-- HDFS Integration -->
  <property>
    <name>hbase.rootdir</name>
    <value>hdfs://mycluster/hbase</value>
  </property>

  <!-- ZooKeeper -->
  <property>
    <name>hbase.zookeeper.quorum</name>
    <value>10.X.X.2, 10.X.X.1, 10.X.X.3</value>
  </property>

  <!-- Cluster Mode -->
  <property>
    <name>hbase.cluster.distributed</name>
    <value>true</value>
  </property>

  <!-- Master/Backup -->
  <property>
    <name>hbase.master</name>
    <value>10.X.X.1:60000</value>
  </property>
  <property>
    <name>hbase.backup.masters</name>
    <value>10.X.X.2</value>
  </property>

  <!-- Network -->
  <property>
    <name>hbase.master.info.port</name>
    <value>16010</value>
  </property>
  <property>
    <name>hbase.regionserver.info.port</name>
    <value>16030</value>
  </property>
</configuration>
```

2. sudo vim/home/sysadmin/hbase-2.3.5/conf/regionservers
10.X.X.3

3. sudo vim/home/sysadmin/hbase-2.3.5/conf/backup-masters
10.X.X.2

```
hdfs dfs -mkdir /hbase
hdfs dfs -ls /

vim  ~/.bashrc

export HBASE_HOME=/home/sysadmin/hbase-2.3.5
export HBASE_CONF_DIR=/home/sysadmin/hbase-2.3.5/conf
export HBASE_REGIONSERVERS=/home/sysadmin/hbase-2.3.5/conf/regionservers
export HBASE_BACKUP_MASTERS=/home/sysadmin/hbase-2.3.5/conf/backup-masters
export HBASE_LOG_DIR=$HBASE_HOME/logs

source  ~/.bashrc

scp -r   hbase-2.3.5 sysadmin@10.X.X.1:/home/sysadmin/
scp -r   hbase-2.3.5 sysadmin@10.X.X.2:/home/sysadmin/


// Start HBase Services
** On 10.X.X.2 (HBase Master):
hbase-daemon.sh start master
hbase-daemon.sh start master --backup       on 10.X.X.2
hbase-daemon.sh start regionserver          on 10.X.X.3

jps

/usr/local/hbase/bin/start-hbase.sh
/usr/local/hbase/bin/stop-hbase.sh


// HBase Shell:
/usr/local/hbase/bin/hbase shell

Check failover When Master stop then Backup-Master become Master
// on 10.X.X.1 Master
hbase-daemon.sh stop master

http://10.X.X.1:16010/
http://10.X.X.2:16010/
```

Not secure 10.13.10.217:16010/master-status

New Chrome available

All Bookmarks

APACHE HBASE

Home | Table Details | Procedures & Locks | HBCK Report | Process Metrics | Local Logs | Log Level | Debug Dump | Metrics Dump | Profiler | HBase Configuration

Master master1

Region Servers

Base Stats | Memory | Requests | Storefiles | Compactions | Replications

ServerName	Start time	Last contact	Version	Requests Per Second	Num. Regions
worker1,16020,1742969211210	Wed Mar 26 11:36:51 IST 2025	0 s	2.3.5	0	2
Total:1				0	2

Backup Masters

ServerName	Port	Start Time
master2	16000	Wed Mar 26 11:53:33 IST 2025

10.13.10.217:16010/processMaster.jsp

Not secure 10.13.10.213:16010/master-status

New Chrome available

All Bookmarks

APACHE HBASE

Home | Table Details | Process Metrics | Local Logs | Log Level | Debug Dump | Metrics Dump | Profiler | HBase Configuration

Backup Master master2

Current Active Master: master1

Tasks

Show All Monitored Tasks | Show non-RPC Tasks | Show All RPC Handler Tasks | Show Active RPC Calls | Show Client Operations

View as JSON

Start Time	Description	State	Status
Wed Mar 26 11:53:37 IST 2025	Master startup	RUNNING (since 34sec ago)	Another master is the active master, master1,16000,1742969562353; waiting to become the next active master (since 34sec ago)

Software Attributes

Attribute Name	Value	Description
JVM Version	Private Build 1.8.0_442-25.442-b06	JVM vendor and version

Not secure 10.13.10.212:16030/rs-status

New Chrome available

All Bookmarks

APACHE HBASE

Home | Process Metrics | Local Logs | Log Level | Debug Dump | Metrics Dump | Profiler | HBase Configuration

RegionServer worker1,16020,1742969211210

Server Metrics

Base Stats | Memory | Requests | WALs | Storefiles | Queues | ByteBuffAllocator Stats

Requests Per Second	Num. Regions	Block locality	Block locality (Secondary replicas)	Slow WAL Append Count
0	2	100.000%	0.000%	0

Block Cache

Base Info | Config | Stats | L1 | L2

Attribute	Value	Description
Implementation	LruBlockCache	Block cache implementing class

See block cache in the HBase Reference Guide for help.

Step-by-Step Guide to Set Up Hadoop Single-Node Cluster

1. Prerequisites
Ensure that you have:
- A Linux-based OS (Ubuntu, CentOS, etc.)
- Java installed (Java 8 or Java 11)
- SSH access configured for the hadoop user

2. Install Java (if not already installed)
```
sudo apt update -y
sudo apt install openjdk-11-jdk -y
java -version
```

3. Add Hadoop User
```
sudo useradd -m -s /bin/bash hadoop
echo "hadoop:hadoop" | sudo chpasswd
sudo usermod -aG sudo hadoop
su - hadoop
```

4. Configure Passwordless SSH
```
ssh-keygen -t rsa -P '' -f ~/.ssh/id_rsa
cat ~/.ssh/id_rsa.pub >> ~/.ssh/authorized_keys
chmod 600 ~/.ssh/authorized_keys
ssh localhost   # Verify that SSH works without a password
```

5. Download and Extract Hadoop
```
cd /home/hadoop
wget https://archive.apache.org/dist/hadoop/core/hadoop-3.1.1/hadoop-3.1.1.tar.gz
tar -xvzf hadoop-3.1.1.tar.gz
```

6. Set Hadoop Environment Variables
vim /home/hadoop/.bashrc and add:

```
export HADOOP_HOME=/home/hadoop/hadoop-3.1.1
export PATH=$PATH:$HADOOP_HOME/bin:$HADOOP_HOME/sbin
export JAVA_HOME=/usr/lib/jvm/java-11-openjdk-amd64
export HDFS_NAMENODE_USER=hadoop
export HDFS_DATANODE_USER=hadoop
export HDFS_SECONDARYNAMENODE_USER=hadoop
export HADOOP_CONF_DIR=$HADOOP_HOME/etc/hadoop

source /home/hadoop/.bashrc
```

7. Create Hadoop Data Directories

```
mkdir -p ~/hadoop_data/namenode
mkdir -p ~/hadoop_data/datanode
chown -R hadoop:hadoop ~/hadoop_data
chmod -R 755 ~/hadoop_data
```

8. Configuration Files

vim /home/hadoop/hadoop-3.1.1/etc/hadoop/core-site.xml

```
<configuration>
   <property>
      <name>fs.defaultFS</name>
      <value>hdfs://localhost:9000</value>
   </property>
</configuration>
```

vim /home/hadoop/hadoop-3.1.1/etc/hadoop/hdfs-site.xml

```
<configuration>
   <property>
      <name>dfs.replication</name>
      <value>1</value>
   </property>
   <property>
      <name>dfs.namenode.name.dir</name>
      <value>/home/hadoop/hadoop_data/namenode</value>
   </property>
   <property>
      <name>dfs.datanode.data.dir</name>
      <value>/home/hadoop/hadoop_data/datanode</value>
   </property>
</configuration>
```

vim /home/hadoop/hadoop-3.1.1/etc/hadoop/yarn-site.xml

```
<configuration>
   <property>
      <name>yarn.nodemanager.aux-services</name>
      <value>mapreduce_shuffle</value>
   </property>
   <property>
      <name>yarn.resourcemanager.hostname</name>
      <value>localhost</value>
   </property>
</configuration>
```

vim /home/hadoop/hadoop-3.1.1/etc/hadoop/mapred-site.xml

```
<configuration>
   <property>
      <name>mapreduce.framework.name</name>
      <value>yarn</value>
   </property>
</configuration>
```

vim /home/hadoop/hadoop-3.1.1/etc/hadoop/hadoop-env.sh

export HADOOP_PREFIX=/home/hadoop/hadoop-3.1.1
export PATH=$PATH:$HADOOP_PREFIX/bin
export JAVA_HOME=/usr/lib/jvm/java-11-openjdk-amd64
export HDFS_NAMENODE_USER=hadoop
export HDFS_DATANODE_USER=hadoop
END ANSIBLE MANAGED BLOCK
Set Hadoop heap size (optional, adjust as needed)
export HADOOP_HEAPSIZE=1024

Set Hadoop user for running services
export HDFS_NAMENODE_USER=hadoop
export HDFS_DATANODE_USER=hadoop
export HDFS_SECONDARYNAMENODE_USER=hadoop
export YARN_RESOURCEMANAGER_USER=hadoop
export YARN_NODEMANAGER_USER=hadoop

source /home/hadoop/hadoop-3.1.1/etc/hadoop/hadoop-env.sh

cd /home/hadoop/hadoop-3.1.1
hdfs namenode -format

cd sbin
./start-dfs.sh
./start-yarn.sh

jps

10. Verify Hadoop Web UI
NameNode UI: http://localhost:9870
Resource Manager UI: http://localhost:8088

##

Steup HBASE-->

1. Download and Extract 2.3.5
cd /home/hadoop
wget https://archive.apache.org/dist/hbase/2.3.5/hbase-2.3.5-bin.tar.gz
tar -xvzf hbase-2.3.5-bin.tar.gz

2. Configure Environment
Edit the -env.sh file:
vim /home/hadoop//conf/-env.sh

export JAVA_HOME=/usr/lib/jvm/java-11-openjdk-amd64
export _MANAGES_ZK=false # Set to true if you want to manage ZooKeeper
export _HOME=/home/hadoop/hbase-2.3.5
export PATH=$PATH:$_HOME/bin

3. Configure hbase-site.xml
Edit the HBase configuration file:
vim /home/hadoop/hbase-2.3.5/conf/hbase-site.xml

```
<configuration>
   <property>
      <name>hbase.rootdir</name>
      <value>hdfs://localhost:9000/hbase</value>
   </property>
   <property>
      <name>hbase.cluster.distributed</name>
      <value>true</value>
   </property>
   <property>
      <name>hbase.zookeeper.quorum</name>
      <value>localhost</value>
   </property>
</configuration>
```

4. Configure regionservers
echo "localhost" > /home/hadoop/hbase-2.3.5/conf/regionservers

5. Start HBase
cd /home/hadoop/hbase-2.3.5/bin
./start-hbase.sh

6. Verify HBase
jps # You should see HMaster, HRegionServer, and potentially ZooKeeper if it's running.
hbase shell

##

zookeeper setup -->

1. Download and Extract ZooKeeper 3.4.6

cd /home/hadoop
wget https://archive.apache.org/dist/zookeeper/zookeeper-3.4.6/zookeeper-3.4.6.tar.gz
tar -xvzf zookeeper-3.4.6.tar.gz

2. Configure ZooKeeper
mkdir -p /home/hadoop/zookeeper-3.4.6/data

Edit zoo.cfg
vim /home/hadoop/zookeeper-3.4.6/conf/zoo.cfg

tickTime=2000
dataDir=/home/hadoop/zookeeper-3.4.6/data
clientPort=2181

```
initLimit=5
syncLimit=2
server.1=localhost:2888:3888

##If you are setting up a multi-node ZooKeeper cluster, you need to add server.2, server.3, etc.
```

3. Set Up ZooKeeper Environment
Edit the /home/hadoop/.bashrc file:

```
vim /home/hadoop/.bashrc

export ZOOKEEPER_HOME=/home/hadoop/zookeeper-3.4.6
export PATH=$PATH:$ZOOKEEPER_HOME/bin

source ~/.bashrc
```

4. Start ZooKeeper

```
cd /home/hadoop/zookeeper-3.4.6/bin
./zkServer.sh start

./zkServer.sh status
```

5. Verify ZooKeeper

```
zkCli.sh
```

Inside the ZooKeeper CLI, run:

```
ls /

# If ZooKeeper is working, it should return [] (empty list) or existing nodes.
```

6. Start HBase with ZooKeeper

```
cd /home/hadoop/hbase-2.3.5/bin
./start-hbase.sh

jps
```

##

1. Uninstall Hadoop

```
# Remove Hadoop binaries
sudo rm -rf /home/hadoop/hadoop-3.1.1
sudo rm -rf /home/hadoop/hadoop_data
sudo deluser -f hadoop
sed -i '/HADOOP_HOME/d' ~/.bashrc
sed -i '/JAVA_HOME/d' ~/.bashrc
sed -i '/HADOOP_CONF_DIR/d' ~/.bashrc
source ~/.bashrc
```

2. Uninstall HBase

```
# Remove HBase binaries
sudo rm -rf /home/hadoop/hbase-2.3.5
sudo deluser hadoop
hdfs dfs -rm -r /hbase
sed -i '/HBASE_HOME/d' ~/.bashrc
source ~/.bashrc
```

3. Uninstall ZooKeeper

```
sudo rm -rf /home/hadoop/zookeeper-3.4.6
sudo deluser hadoop
sudo rm -rf /home/hadoop/zookeeper-3.4.6/data
sed -i '/ZOOKEEPER_HOME/d' ~/.bashrc
source ~/.bashrc

sudo apt-get autoremove
sudo apt-get clean
```

##